

HERITAGE Family Study Review Paper

Supplementary Tables

Supplementary Table S1. HERITAGE Consortium Steering Committee

Claude Bouchard, PhD; <i>Chairperson</i>	Laval University (until 1999); later Pennington Biomedical Research Center
Jacques Gagnon, PhD; <i>Project Director</i> (1992-1999)	Laval University
Jean Paul Albert, MBA; <i>Administrator</i> (1992-1999)	Laval University
Tuomo Rankinen, PhD; <i>Project Director</i> (1999-2012)	Pennington Biomedical Research Center
Arthur S. Leon, MD	University of Minnesota
D. C. Rao, PhD	Washington University
James S. Skinner, PhD	Arizona State University (until 1995); Indiana University (1996+)
Jack H. Wilmore, PhD	The University of Texas at Austin (until 1997), later Texas A&M University, then back to The University of Texas at Austin after 2003.

Supplementary Table S2. HERITAGE Consortium Advisory Board (1992-2008)

Elizabeth Barrett-Connor, MD	University of California, San Diego
Jean Davignon, MD	Clinical Research Institute of Montreal
E. Randy Eichner, MD (1992-2002)	University of Oklahoma
Robert C. Elston, PhD	Louisiana State University and Case Western Reserve University
William L. Haskell, PhD	Stanford University
Rudy Leibel, MD (2002-2008)	Columbia University
Eric Ravussin, PhD (2002-2008)	Pennington Biomedical Research Center

Supplementary Table S3. HERITAGE Coordinating Centers, Clinical Centers, and Core Laboratories: List of Personnel

Laval University Consortium Coordinating Center and Clinical Center

PI: Claude Bouchard, PhD

Local Project Coordinator: Marcelle Lareau, MSc

Jean-Paul Albert, MBA

Manon Baril, BSc

Jean Bergeron, MD

Come S. Bouchard, MSc

Anne-Marie Bricault, MSc

Yvon Chagnon, PhD

Jean-Pierre Despres, PhD

Diane Drolet, MSc

Jacques Gagnon, PhD

My-Anh Ho-Kim, MSc

Karen Horth,

Louise Laberge,

Benoit Lamarche, PhD

Claude Leblanc, MSc

Gilles Lortie, MD, PhD

Isabel Mercier, MSc

François Michaud, BSc

Chantal Paré, BSc

Louis Pérusse, PhD

Denis Prud'homme, MD, MSc

Jean-Aime Simoneau, PhD

Germain Thériault, MD

Angelo Tremblay, PhD

John Weisnagel, MD, PhD

Washington University Data Coordinating Center

PI: D. C. Rao, PhD

Ping An, MD

Ingrid B. Borecki, PhD

Harry Cheng, MSc

Gu Chi, PhD

Warwick Daw, PhD

Habib El-Moalem, PhD

John O. Holloszy, MD

Yuling Hong, MD

David J. Lerner, BSc

Zhaohai Li, PhD

Stephen Mendel, PhD

Laura E. Mitchell, PhD

Michael A. Province, PhD

Treva Rice, PhD

Kathryn A. Schallert, BSc

Kenneth B. Schechtman, PhD

George P. Vogler, PhD

Laval University Core Laboratories

Diabetes Research Unit

André Nadeau, MD, PhD; Director
Yolande Montreuil, RN
Gilles Tancrede, MSc

Lipid Research Center

Paul Lupien, MD, PhD; Director
Jean-Pierre Després, PhD
Sital Moorjani, PhD

DNA and Cell Line Unit

France T. Dionne, PhD; Director
Monique Chagnon, ART
Caroline Noel, BSc
Marie-Christine Thibault, PhD

Steroid Laboratory

Alain Bélanger, PhD; Director
Simon Caron, MSc
Fernand Labrie, MD, PhD
Line Lavoie, MSc

University of Minnesota Clinical Center

PI: Arthur S. Leon, MD
Local Project Coordinator: Ava J. Walker, PhD

Marilyn Borkon,
Fernando S. Branco, MD
Erik J. Ekstrom, RD
Steve Gaskill, PhD
William V. Mendoza, MD

Marcella Meyers, PhD
James P. Norton, MSc
Robert C. Serfass, PhD
M. Katie Schmitz, MSc

Arizona State University Clinical Center (Until 1995)

Indiana University Clinical Center (1996+)

PI: James S. Skinner, PhD
Local Project Coordinator in Arizona: Kristine M. Wilmore, MA
Local Project Coordinator in Indianapolis: Joanne Krasnoff, PhD

Kelly Enders, RN
Anna A. Jaskolska, PhD
Artur J. Jaskolski, PhD

University of Texas at Austin Clinical Center

PI: Jack H. Wilmore, PhD

Local Project Coordinator: Philip R. Stanforth, MSc

Melissa Domenick, MA

David W. Hayes, MD

Paul J. Roach, MD

Supplementary Table S4. Listing of some of the most novel or impactful findings from the HERITAGE Family Study from 1995 to 2021.

Novel or Impactful Findings	Trait(s)	Details	Citation(s)
Heterogeneity of Training Responses			
Large inter-individual differences in phenotype responses to endurance training	VO ₂ max	VO ₂ max responses ranged from no gain to over 1 L/min [-5% to +48%]	Bouchard and Rankinen 2001; Skinner, Jaskolski et al. 2001
	IVGTT traits	The proportion of subjects who had no change or decreases in IVGTT traits ranged from 42-55%	Boule, Weisnagel et al. 2005
No correlation between baseline VO ₂ max and its changes with training	VO ₂ max	The correlation between baseline and delta VO ₂ max was 0.08, with baseline levels after adjustments explaining only 2% of the variance in delta VO ₂ max	Skinner, Jaskolski et al. 2001; Sarzynski, Ghosh et al. 2017
No aggregation of training responsiveness in individuals or subgroups	VO ₂ max	There were low, average, and high VO ₂ max responses across ages, both sexes, both ethnic groups, and levels of initial VO ₂ max.	Skinner, Jaskolski et al. 2001
	VO ₂ max, %fat, AVF, insulin, HDL-C, small LDL, GlycA	Almost half of the cohort had at least one high response and one low response across seven cardiometabolic traits. There is a high degree of trait specificity in training responsiveness.	Barber BJSM in press

Some unfavorable responses to endurance training	fasting insulin; HDL-C; TG; SBP	Prevalence of potentially "adverse" response in HERITAGE ranged from 6 to 9% for the four traits.	Bouchard Plos One 2012
Familial Aggregation of Baseline Phenotype Levels			
Within-subject variability	VO ₂ max	Within-subject standard deviation from measures repeated days and weeks apart range from 108 to 137 mL O ₂ /min with CV of 4.1 to 5.0%	Wilmore, Stanforth et al, 1998; Skinner, Wilmore et al, 1999; Shephard, Rankinen et al, 2004; Sarzynski, Ghosh et al, 2017; plus unpublished data.
Significant familial resemblance for intrinsic VO ₂ max	VO ₂ max	2.7 times more variance between families than within families. Heritability of intrinsic VO ₂ max was 51%, with significant maternal heritability (29%)	Bouchard, Daw et al. 1998
Significant familial aggregation for intrinsic submaximal VO ₂	VO ₂ at 50W, 60%, 80%	Maximal heritability estimates ranged from 48-70% and 29-48% for maternal heritability	Perusse, Gagnon et al. 2001
Significant familial aggregation for intrinsic submaximal Q & SV	SV & Q at 50W and 60% VO ₂ max	Maximal heritability estimates of 46% for SV and Q at 60% VO ₂ max and 41% & 42% at 50W	An, Rice et al. 2000
Significant familial aggregation for visceral fat	AVF	Heritability of 47-48%, independent of total body fat	Rice, Despres et al. 1997
Higher heritability of resting BP in Black subjects	Resting SBP & DBP	Heritability of SBP & DBP were 68% and 56% in Black subjects vs 43% & 24% in White subjects.	Gu 1998

Significant familial aggregation for muscle enzyme activities	CK, PHOS, HK, PFK, GAPDH, LPL, CPT, HADH, CS, COX	Strong familial aggregation for activities of baseline muscle enzymes related to PCr, glycolytic, and oxidative metabolism.	Rico-Sanz, Rankinen et al. 2003; Rankinen, Bouchard et al. 2005
Significant familial aggregation for markers of oxidative stress	LDL-ox, C50-AAPH, TBARS, glutathione	Heritability for oxidative stress traits ranged from 31% to 44%.	Blache, Lussier-Cacan et al. 2007
Familial Aggregation of Training Response			
Significant familial aggregation for VO ₂ max training response	VO ₂ max	Heritability was 47%, with maternal inheritance accounting for 28%	Bouchard, An et al. 1999
Significant familial aggregation for submaximal VO ₂ training response	VO ₂ at 50W, 60%, 80%	Heritability values ranged from 23% to 57% for the training response of submaximal measures of VO ₂	Perusse, Gagnon et al. 2001
Significant familial aggregation of submaximal exercise BP and HR training responses	HR50, HR60%, SBP50	Heritability for HR50, HR60%, SBP50 responses were 34%, 29%, 22%. Heritability for DBP traits and all HR and BP traits were lower in Black subjects	An, Perusse 2003
Significant familial aggregation for submaximal Q & SV training responses	SV & Q at 50W and 60% VO ₂ max	Heritability ranging from 24 to 38% for SV and Q at 50W and 60% VO ₂ max	An, Rice et al. 2000
Significant familial aggregation of inter-individual variation in plasma lipid responsiveness to training	TC, TG, LDL-C, apoB, HDL-C, HDL2-C, HDL3-C, apoA-I	Heritability ranged from 25% to 38% for lipid response traits. Exceptions were for heritability levels near 60% for changes in apoB in Blacks and HDL2-C in Whites and a lack of heritability for change in LDL-C in Black subjects	Rice, Despres et al. 2002
Significant familial aggregation for training response of markers of oxidative stress	LDL-ox, C50-AAPH, TBARS, glutathione	Heritability for oxidative stress training response traits ranged from 35% to 84%	Blache, Lussier-Cacan et al. 2007

Significant familial aggregation for training response of muscle enzyme activities	CK, PHOS, HK, PFK, GAPDH, CPT, HADH, CS, COX	Strong familial aggregation found for training response of muscle enzymes related to PCr, glycolytic, and oxidative metabolism	Rico-Sanz, Rankinen et al. 2003; Rankinen, Bouchard et al. 2005
Ethnic and sex differences in responses to training			
Training responses of submaximal exercise measures of hemodynamic traits differed by sex and ethnic groups	HR, SBP, DBP at 50W	Submaximal HR, SBP, DBP decreased with training, with greater reductions in women compared to men and in Black and older subjects compared to White and younger subjects.	Wilmore, Stanforth 2001 pp 10-116
	a-vO ₂ diff, SV, Q, and VO ₂ at 50W	Black men did not increase a-vO ₂ diff at 50W. Thus, on average Black men had greater increases in SV50 and smaller decreases in Q50 compared with White men to achieve similar VO250.	Wilmore, Stanforth 2001 MSSE pp 99-106
Significant sex interactions for insulin sensitivity response to training	Si	Insulin sensitivity increased by 10% in the total cohort, with the increase larger in men than women (16% vs 5%).	Boule, Weisnagel et al. 2005
Significant ethnic and sex differences in lipid, lipoprotein, and lipase activity responses to training	ApoA-I, HDL2-C	ApoA-I increased more in women than men, in Black than White subjects, and in offspring than in parents. Black subjects experienced greater increases in HDL2-C compared to White subjects.	Leon, Rice et al. 2000
	LPL activity	LPL activity increased in all subgroups except Black men.	Bergeron, Couillard et al. 2001
Response of other traits to training			
Little influence of endurance training on markers of oxidative stress	LDL-ox, C50-AAPH, TBARS, antioxidants, and aminothiols	Only erythrocyte resistance to hemolysis significantly changed with training, which interacted with smoking status (smokers did not experience beneficial effects of training on erythrocytes), and was significant in women only.	Blache, Lussier-Cacan et al. 2007

No change in RMR with training .	RMR via indirect calorimetry	Sample size (N=77) was larger than previous studies. No change in RMR at 24 or 72 hrs post-training.	Wilmore, Stanforth et al. 1998
Favorable changes in clinically relevant lipoprotein subfractions in response to training	NMR-based lipoprotein subfractions	Large HDL-P and LDL-P increased, while small LDL-P and all VLDL subfractions and VLDL-P size decreased with training. These findings were not captured with traditional lipid profiling (i.e., TG and LDL-C did not change in total sample).	Sarzynski, Burton et al. 2015

Genome-wide linkage studies of baseline and response phenotypes

Identification of QTLs for baseline VO ₂ max and for VO ₂ max response	VO ₂ max	QTLs on 4q, 8q, 11p, and 14q reported for baseline VO ₂ max. QTLs on 1p, 2p, 4q, 6p, and 11p identified for change in VO ₂ max.	Bouchard, Rankinen et al. 2000
Dense mapping of four QTLs identified strong candidate genes for submaximal exercise capacity and hemodynamic responses to training	SV50, HR50	Titin, kinesin family member 5B, cAMP responsive element binding protein 1, <i>MIPEP</i> and <i>SGCG</i> genes identified as strong candidates for changes in submaximal SV and HR and submaximal exercise capacity.	Rankinen, Rice et al. 2003; Argyropoulos, Stutz et al. 2009; Rankinen, Argyropoulos et al. 2010; Rice, Sarzynski et al, 2012

First GWAS of exercise response traits

GWAS of VO ₂ max response to endurance training	VO ₂ max	39 SNPs were associated at $p < 1.5 \times 10^{-4}$, with a panel of 21 SNPs accounting for 49% of the variance in VO ₂ max trainability.	Bouchard, Sarzynski et al. 2011
GWAS of submaximal HR response to training	HR50	40 SNPs associated at $p < 9.9 \times 10^{-5}$, with top hit 8×10^{-7} . Nine SNPs accounted for the genetic variance of the submaximal exercise HR response to training.	Rankinen, Sung et al. 2012

Molecular signatures of exercise response derived from integrative analyses of genomic and transcriptomic profiles

Muscle gene expression and SNP signatures predict VO ₂ max response to endurance training	VO ₂ max	Genome-wide baseline muscle gene expression and validation identified a 29-RNA signature that predicted changes in VO ₂ max. Candidate genes from this predictor and the literature led to a 11	Timmons, Knudsen et al. 2010
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SNP signature that explained 23% of the variance in VO₂max trainability.

Combined genome-wide and transcriptome-wide analysis identifies SNPs associated with TG response to training	TG	GWAS identified 4 SNPs accounting for the genetic variance of TG response, while molecular signature based on the baseline expression of 11 genes predicted 27% of TG changes in response to training. An 8-SNP score comprised of 4 SNPs each from transcriptomics and GWAS was the strongest predictor of TG training response.	Sarzynski, Davidsen et al. 2015
GWAS and transcriptional signature of insulin sensitivity response to training	Si	Integrative analysis of functional genomic and transcriptomic profiles identified combined variation in genes linked to cholinergic, calcium, and chemokine signaling associated with Si training response. MEF2A transcription factor was the most significant candidate driving the transcriptional signature associated to ΔSi, strengthening the relevance of calcium signaling in exercise training-mediated Si response.	Takeshita, Davidsen et al. in press

Proteomic signatures of VO₂max and its trainability

Plasma protein signature of cardiorespiratory fitness level	VO ₂ max	Elastic net regression identified 115 proteins highly correlated (r ² =0.80) with measured VO ₂ max, which was replicated in the validation dataset, with an r ² of 0.71 (Figure 9).	Williams, Kivimaki et al. 2019
Plasma proteins associated with intrinsic VO ₂ max and its trainability	VO ₂ max	147 proteins were associated with baseline VO ₂ max, while 102 baseline proteins were associated with changes in VO ₂ max, with minimal overlap (only 5 proteins) between protein sets. A baseline 56 protein signature improved prediction of VO ₂ max response (AUC 0.84) compared to a model of only clinical variables (AUC 0.62).	Robbins, Peterson et al. 2021

Bioinformatics explorations of intrinsic VO₂max and its trainability

Integrative pathway analysis of VO ₂ max response to training GWAS	VO ₂ max	Using GWAS results followed by candidate gene prioritization and pathway analysis, pathways related to calcium signaling, energy sensing and partitioning, mitochondrial biogenesis, angiogenesis, immune functions, and regulation of autophagy and apoptosis were identified as key mechanisms through which the physiological responses of VO ₂ max to training are mediated.	Ghosh, Vivar et al. 2013
Genetics and biology underlying intrinsic VO ₂ max	VO ₂ max	A bioinformatics pipeline applied to VO ₂ max data in the sedentary state suggests four loci related to cardiovascular physiology (<i>ATE1</i> , <i>CASQ2</i> , <i>NOTO</i> , and <i>SGCG</i>), four loci related to hematopoiesis (<i>PICALM</i> , <i>SSB</i> , <i>CASQ2</i> , and <i>CA9</i>), four loci related to skeletal muscle function (<i>SGCG</i> , <i>DMRT2</i> , <i>ADARB1</i> , and <i>CASQ2</i>), and eight loci related to metabolism (<i>ATE1</i> , <i>PICALM</i> , <i>RAB11FIP5</i> , <i>GBA2</i> , <i>SGCG</i> , <i>PRADC1</i> , <i>ARL6IP5</i> , and <i>CASQ2</i>) as candidates for human variability in cardiorespiratory fitness among sedentary adults.	Ghosh, Hota et al. 2019

Metabolomic biomarker of training responsiveness

DMGV is a biomarker of metabolic response to endurance training	Fasting glucose, insulin and lipids	Baseline levels of plasma DMGV associated with lack of improvement in HDL traits. DMGV levels decreased with training and were positively correlated with several lipid, glucose and insulin traits	Robbins, Herzig et al, 2019
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